Gunnar Arias



Computer Science student (BSc, June 2027) · Data, ML, fintech & backend engineering

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PROFILE

Final-year Computer Science student (grade average 4.0/5) who builds typed, tested backend services and data pipelines in Python, TypeScript and Kotlin, with a focus on correctness: double-entry ledgers, idempotent APIs, concurrency control and leakage-safe feature engineering. Hackathon winner. Swedish (EU) citizen, open to roles across the EU/EEA; graduating June 2027 and looking for internships and graduate/junior roles in ML, data, fintech/banking tech and football analytics.

SELECTED PROJECTS

Ledger Forge API | *Kotlin, Spring Boot, PostgreSQL, Flyway, Docker Compose, OpenAPI* github.com/gunnarwrld/ledger-api 2026

- Double-entry wallet ledger storing money as integer minor units (ISO 4217); balanced journal entries validated in the service and re-checked at commit by a deferred PostgreSQL constraint trigger.
- Idempotent transfers (Idempotency-Key + SHA-256 request fingerprint, 409 on conflicting reuse); ordered SELECT … FOR UPDATE row locks prevent double-spending; append-only audit trail; unit/integration tests in CI.

Forge Wallet (Android, in progress) | *Kotlin, Jetpack Compose, Room, WorkManager* github.com/gunnarwrld/forge-wallet-android 2026

- Offline-first wallet on Ledger Forge: payments queue in a Room outbox, a background sync worker survives airplane mode and process death, and idempotency keys make each transfer settle exactly once; payment path works end to end.

Fútbol Data Aggregator API | *TypeScript, Express 5, PostgreSQL, Redis, BullMQ* github.com/gunnarwrld/futbol_data_aggregator_api 2026

- Live service on Railway that ingests live and historical football data from API-Football with scheduled BullMQ workers, normalises it into PostgreSQL and serves versioned REST endpoints plus a Socket.io real-time channel.
- Redis caching, sliding-window rate limiting and API-key auth; layered architecture with Vitest tests, ESLint and Husky hooks.

Fair Football Fee (in progress) | *Python, pandas, DuckDB, Parquet, pytest* github.com/gunnarwrld/fair-football-fee 2026

- Player market-value predictor for Premier League players: data audit, raw ingestion to Parquet, cleaning and strictly point-in-time feature engineering, with tests for data leakage and each pipeline phase; modelling in progress.

Football Score Prediction (ML course project) | *Python, NumPy, scikit-learn* 2026 – ongoing

- Group project predicting home and away scores of CONMEBOL World Cup qualifiers from pre-match features; linear regression implemented from scratch in NumPy and compared with a scikit-learn Random Forest regressor.

HKIF Sports Union Platform | *Express 5, TypeScript, Prisma, PostgreSQL, JWT, Vitest, Supertest* hkif-web.vercel.app 2025

- Built the REST API from scratch with a typed error/response contract, JWT auth, 4-tier role-based access control and DTO projection; set up the Vitest + Supertest integration suite.

Also: Clicky Recurring Runs (Python, FastAPI, Pydantic, pytest; duplicate-safe scheduled runs) · FinanceTracker PWA (React, TypeScript, Supabase; multi-currency, Google sign-in, offline support) · AI Study Assistant (Cloudflare Workers, Durable Objects, Llama 3.3 70B, WebSockets) · gunnarwrld.com (React, Vercel Functions, Vitest, Playwright, CI).

AWARDS

Winner, "Shaping the City of the Future" hackathon | *Team City Devs · Kristianstad Univ. & COLOURS Alliance (EU co-funded)* Sep 2026

EXPERIENCE

Shift Leader (Driftledare), Max Burgers, Sweden | *Part-time* May 2026 – Present

- Lead shifts alongside full-time studies: coordinate the team and day-to-day operations, and handle customer issues as the person in charge.

Youth Football Coach (volunteer), Club Castillo de Amaguaña, Ecuador | *Coached and led a children's amateur football team*

EDUCATION

BSc Computer Science, Kristianstad University, Sweden Sep 2024 – expected Jun 2027

Grade average 4.0 / 5 (Swedish scale 3–5); 124 credits completed. Distinction (5) in OOP in Java, Discrete Mathematics, Computer Security, Data Communication, Full Stack Development, Intro to CS. Also: Mathematical Statistics, Database Technique, Algorithms & Data Structures; ongoing: Machine Learning, Software Engineering, Mobile Development.

Certificate in Digital Marketing, Linköping University, Sweden Sep 2023 – Jul 2024

SKILLS & LANGUAGES

Programming: Python, SQL, TypeScript, JavaScript, Kotlin, Java, HTML, CSS
Data & ML: pandas, NumPy, scikit-learn, DuckDB, PyArrow/Parquet; regression modelling, feature engineering, data validation
Backend: Spring Boot, FastAPI, Node.js, Express, PostgreSQL, MySQL, SQLite, Redis, BullMQ, Prisma, Supabase, REST, OpenAPI
Mobile & frontend: Android (Jetpack Compose, Room, WorkManager, Retrofit), React, Vite, Tailwind; LLM integration (Llama 3.3)
Testing & tools: pytest, Vitest, Supertest, Playwright; Git, GitHub Actions, Docker Compose, Vercel, Railway, Cloudflare Workers
Languages: Spanish (native) · English (fluent) · Portuguese (intermediate) · Swedish (intermediate, B1)